BATCH 29: NBA – MVP Prediction

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1. What the Stanford project did

The Stanford CS229 project explored whether statistics could help predict the NBA Most Valuable Player (MVP). It tried two methods: Naive Bayes, reported at about 67% accuracy, and logistic regression, reported at about 50% seasonal accuracy. The project used a fixed train/test split over four seasons and focused on the top three MVP finalists.

In simple terms, Stanford treated the task mainly as choosing the MVP from a small group of finalists. The results showed that player statistics can contain useful signals, but testing on only four seasons gives limited evidence about how well a method will work across many different seasons.

2. What our project did differently

Our project predicts vote share instead of directly predicting a yes/no MVP winner. Vote share is the fraction of all possible MVP voting points received by a player. The model predicts this value for each player in the dataset, then ranks players in a season. The player with the highest predicted share becomes our MVP pick.

We use 32 predictors, including basic box-score statistics, advanced measures such as Win Shares and BPM, and team context such as wins and win percentage. We compare three regression models—Ridge, Random Forest and HistGradientBoosting—with two baselines: prior-season vote share and Ridge using only Win Shares.

Area	Stanford project	Our project
Prediction	MVP winner classification	Predict vote share, then rank
Features	About 10 basic stats	32 stats and team context
Testing	Fixed split; 4 seasons	Walk-forward; 36 test seasons

3. How we tested the models

We use expanding-window walk-forward testing. For each test season, the model trains only on earlier seasons and then predicts that season. The test period is 1990–2025, giving 36 held-out seasons. As time moves forward, the training set grows. This is more realistic than randomly mixing seasons because a prediction should not use information from future seasons.

4. Accuracy comparison

The table below shows the accuracy figures reported in our project README. Winner accuracy means the model selected the actual MVP as its top-ranked player in that season.

Model	Correct picks	Accuracy
Random Forest	25 / 36	69.4%
HistGradientBoosting	24 / 36	66.7%
Ridge Regression	23 / 36	63.9%
Win Shares-only Ridge	21 / 36	58.3%
Prior-season share	10 / 36	27.8%
Stanford Naive Bayes	Not comparable count	~67%
Stanford logistic regression	Not comparable count	~50%

5. What the results mean

Random Forest is the best model in our reported results: it picked the correct MVP in 25 of 36 test seasons, or 69.4%. This is 2.4 percentage points above Stanford's reported Naive Bayes figure of about 67%, and 19.4 points above its reported logistic-regression figure of about 50%. Our Gradient Boosting model is close to the Naive Bayes number at 66.7%.

However, these numbers should be treated as a reference rather than a fair head-to-head contest. Stanford used a different dataset, a smaller candidate group and a different testing setup. So we can say our reported accuracy is numerically higher, but we cannot conclude from these figures alone that our models are definitively better.

6. Limitations and conclusion

Our main improvement is the combination of more predictors and testing across 36 held-out seasons. The source dataset contains only players who received MVP votes, not every player in the NBA, so the model chooses among recorded vote recipients. Also, the reported metrics should be confirmed by running the full notebook and saving its outputs.

Overall, Random Forest performs best among our tested models. The project shows that vote-share regression and walk-forward evaluation are useful ways to study MVP voting, while a stronger direct comparison would require both approaches to be tested on the same seasons and candidate groups.